6 A simple ideal iron-cored transformer is illustrated in Fig. 6.1.

An alternating input p.d. V_p of 240 V is connected across the primary coil, of 120 turns of the transformer. The secondary p.d. V_s of 3600 V is applied to a load resistor.

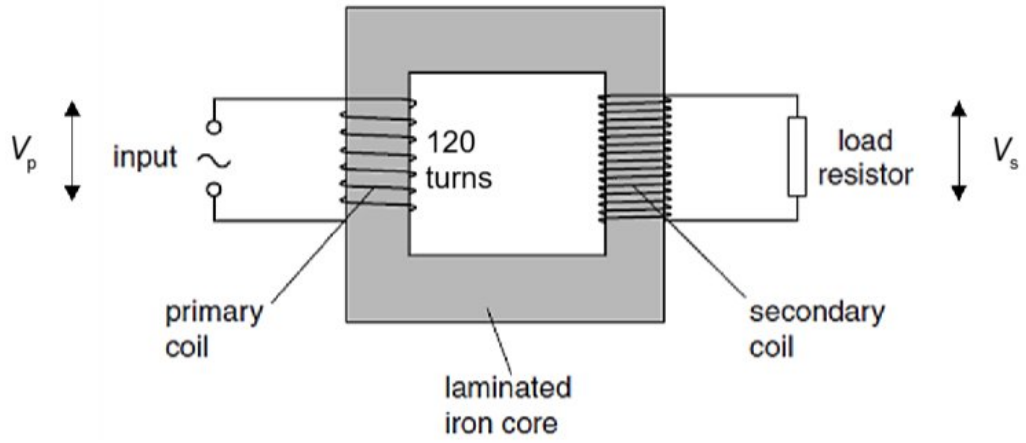


Fig. 6.1

(a) **(i)** Explain why an iron core is preferred over an air core in the construction of a transformer.

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..... [1]

(ii) Explain why the iron core is laminated.

.....

.....

..... [1]

(b) Calculate the number of turns in the secondary coil.

number of turns = ... [1]

(c) Using Faraday's Law, explain why an alternating current in the primary coil gives rise to an alternating e.m.f. in the secondary coil.

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..... [3]

(d) A step-up transformer is used when transmitting electricity over long distances.

Suggest a reason for this.

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..... [1]

[Total: 7]